SECBNet: Semantic Segmentation-Enhanced Color Balance Network for Optical Satellite Images

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***Abstract*— Earth observation satellites can capture optical images under different temporal, climatic conditions, and platforms exhibit substantial differences in color and brightness, leading to poor visual experiences when synthesizing large-area optical satellite images. The related issue of color balancing has attracted considerable attention from researchers, yet challenges such as a lack of research data and sensitivity to model parameters persist. To address these problems, this article publishes a publicly open dataset and presents a semantic segmentation-enhanced color balance network (SECBNet). First, to mitigate the scarcity of research data, we develop a publicly available remote sensing image color balance dataset, Zhu Hai color balance image (ZHCBI), to support related research activities. Second, to improve semantic consistency between the color-balanced images and the target images, we design a dual-branch U-Net architecture guided by segmentation results and propose a novel segmentation feature loss function. Finally, to address issues of seams and unnatural transitions between blocks in segmented processing, we introduce a postprocessing module based on weighted averaging. We conducted comparative experiments and analyses with existing mainstream color balancing algorithms on the ZHCBI dataset. The results demonstrate that our proposed method achieves state-of-the-art color balancing quality, with significant improvement in visual effects and a higher peak signal-to-noise ratio (PSNR) (23.64 dB) compared with other mainstream methods.**

***Index Terms*— Color balance, optical satellite image, seam gap, supervised learning, weighted average algorithm.**

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I. Introduction

DUE to differences in lighting, climate, and imaging platforms, satellite images obtained at different times and from different sources often exhibit significant color discrepancies. These inconsistencies not only lead to poor visual integration but also adversely affect subsequent tasks such as satellite map production [1], [2], road detection [3], [4], and semantic segmentation [5], [6]. The study of color balancing techniques has thus garnered significant attention from scholars [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17]. Current methods for color balancing in remote sensing images can be divided into traditional nondeep learning methods and those based on deep learning.

Traditional image color balancing methods include histogram matching [18], [19], color channel stretching [20], and algorithm optimization [21]. Although these methods are relatively simple to operate, they are typically sensitive to parameter settings and have limited adaptability. With the advancement of deep neural networks, researchers have begun to use convolutional neural networks and transformer architectures to address various image color balancing issues [8], [22], [23], [24]. Compared with traditional methods, those deep learning models are advantageous in terms of processing efficiency and adaptive capacity. However, there are several significant challenges for these methods. First, there is a lack of publicly available datasets for research on color balancing of remote sensing images, which hampers the progression of related studies. Second, due to the large size of remote sensing images, deep learning models typically require the data to be processed in small image blocks, leading to notable color discrepancies and seam issues between different blocks. As illustrated in Fig. 1, which showcases an unprocessed image, an image generated using the TransGAN [8] method, an image produced using our method, and a real ground-truth image, it is evident that the TransGAN-generated image has significant blockwise color differences and seam issues compared with the ground-truth image. Third, the current deep-learning-based methods often result in notable color differences between the same types of targets due to a lack of consideration for semantic consistency.

To tackle these challenges, our article focuses on three main areas. First, we compile a Zhuhai remote sensing image color balance dataset Zhu Hai color balance image

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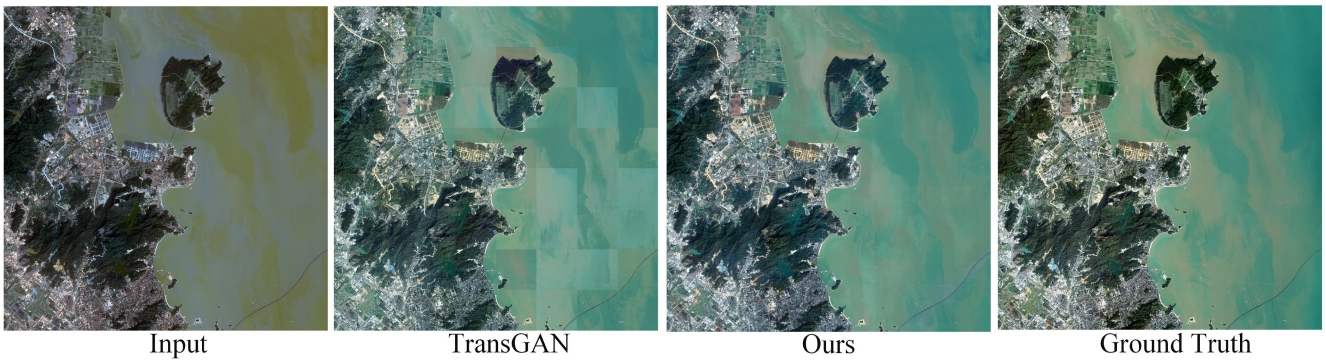


Fig. 1. Comparison of the results of the TransGAN [8] method, our method, and the actual ground-truth image. Our method demonstrates a clear advantage in color balancing and eliminating seam lines, producing images with smoother transitions and more natural color representation, and achieving test results that are more in line with human visual perception standards.

(ZHCBI) from the GF-1 satellite, accessible via this link: http://39.98.109.195:1000/share/xTVn60gY. Second, we introduce a network model that enhances semantic consistency across image blocks using a U-Net-like [25] architecture. This model includes a semantic enhancement module [26], improving feature learning by leveraging a pretrained semantic segmentation network. We also use residual connections to manage parameter explosion and gradient vanishing. Importantly, the semantic segmentation module is used only for feature extraction and not in training. Finally, we propose a postprocessing algorithm based on weighted averaging to address seam issues between image blocks, effectively reducing color differences. Through the proposed method, intelligent and efficient optical satellite image color balance inference can be performed on the data to be processed using an end-to-end network.

II. Related Work

A. Traditional Color Balancing Methods

Traditional image color balancing methods can be broadly categorized into two types: those that directly transform the image and those that use white balance. Direct transformation methods, as presented in [18], [27], and [28], use digital image processing techniques such as Gaussian filtering, histogram equalization, and Wallis transform to adjust image brightness and color differences. In contrast, white balance methods, as demonstrated in [20] and [29], correct color differences by stretching color channels or enhancing color contrast to balance whites in the image. While these traditional methods are user-friendly, they often struggle to gather sufficient prior information for specific scenes, impacting the accuracy of color balancing outcomes.

B. Color Balancing Methods Based on Deep Learning

In contrast to traditional image color balancing methods, deep learning techniques are commonly embraced due to their robust feature learning capabilities. Within the realm of image color balancing, three primary challenges emerge: enhancing color transformations within images, mitigating color discrepancies between generated and real images, and elevating the quality of image color balancing at the semantic perception level.

A triple-state domain translation network was presented in [30] to enhance image color transformations. This network uses two variational autoencoders (VAEs) to map both the images to be processed and real images into separate latent spaces. By doing so, it aims to narrow the gap between the processed and real images' domains, thus learning the mapping relationship between them. Moreover, this network incorporates two local branches to diminish noise and blur, thereby improving its ability to accurately convert colors from processed to real images. Following this, HistoGAN [31] adjusts the color of generated images based on their color histograms. By describing the color characteristics of images visually through histograms and separating their specific color domains, HistoGAN's encoder network is trained to alter colors based on learned histogram priors while preserving the original images' content. In addition, a neural preset technique using a two-stage pipeline was presented in [32], by dividing the task into color normalization and stylization. This method uses color styles as presets to achieve effective color style transformations and minimize visual artifacts in generated images. Finally, MROPM-Net [33] integrates style transfer modules for image stylization and leverages multiple reference images through a data generation approach to produce refined results closer to real images. LLFlow [9] uses an invertible network to map the features of low-light images to a Gaussian distribution, allowing it to effectively learn the complex conditional distribution of properly exposed images. Furthermore, by adopting an illumination-invariant color map as a prior in the latent space, LLFlow retains rich color details while reducing noise and artifacts, thus ensuring the enhanced images exhibit natural brightness and high quality. TransGAN [8] introduces a fully transformer-based generative adversarial network that completely eliminates convolutional layers in both the generator and the discriminator. It uses transformer blocks to gradually increase image resolution and integrates grid self-attention mechanisms to address memory bottlenecks in high-resolution generation.

To tackle the color differences between generated and real images, a deep-learning-based architecture was proposed in [34] for end-to-end image color balancing. This approach

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maps input images into high- and low-illumination domains, and then uses a decoder to merge these domains, effectively eliminating color biases caused by varying scene lighting. Ucolor [35] uses a multicolor space encoder to extract features from diverse color channels, integrating them to enhance feature representation. It further uses a medium transmission decoder network to enhance color expression in degraded image areas, showing remarkable performance. In addition, a combined model was proposed in [36] for color correction, integrating a dehazing module and a correction module. The dehazing module leverages dark channel prior to alleviate the effects of external environments on image color and clarity, while the correction module achieves automatic color correction via low-frequency correction and super-resolution mapping. TOPAL [37] enhances image contrast using a multiscale enhancement module, performs color correction with a deep rendering module, and guides feature extraction for adaptive fusion through a dual-channel attention module, resulting in improved image quality and reduced color disparities. The global feature modeling transformer (GFMT) model [10] presents a U-shaped transformer for image enhancement, incorporating channel-level and spatial-level multiscale feature fusion modules. By focusing on the attenuation characteristics of different color channels and spatial areas, it significantly improves color restoration and contrast. In addition, the network uses a multicolor space loss function, enhancing the color fidelity of the output image to make it more visually consistent with real-life scenes. UNetFormer [17] is a hybrid structured network that adopts ResNet18 as the encoder and a transformer-based decoder, using a global–local attention mechanism to capture both global and local information within the image. The decoder also integrates a feature refinement module to enhance semantic feature resolution and details. MANet [16] is a multiattention network for remote sensing images, using ResNet-50 as its feature extraction backbone. It uses multilevel attention modules, including kernel and channel attention, to effectively capture global context information and adaptively weight feature channels. Compared with traditional attention mechanisms, kernel attention significantly reduces computational complexity while maintaining long-range dependencies, thereby improving fine-grained segmentation accuracy of remote sensing images.

To elevate image color balancing quality at the semantic perception level, DCC-Net [23] uses two decoupling modules to extract grayscale representations and color histograms from images. The grayscale representation generates natural texture structures, while the color histograms ensure color consistency. In addition, they introduce a pyramid color embedding (PCE) module to enhance color information dissemination throughout the balancing process. A multidecoder integrated semantic style transformation model was proposed in [38]. In their model, an encoder furnishes color feature representations to multiple decoders, enhancing color contrast in generated images. This model integrates adaptive instance normalization (AdaIN) with global and local attention modules for precise color calibration, mitigating semantic mismatch. An unsupervised object-based attention mechanism (OBAM) module was proposed in [24] to refine color transfer accuracy and preserve detailed features. The weighted adaptive instance normalization (WAA) method was introduced to compensate for the model's lack of semantic references, enhancing overall perceptual image quality. The semantic-aware knowledge-guided framework (SKF) was proposed in [22], which learns semantic prior knowledge in the feature space through a semantic-aware embedding module. This framework combines color histogram loss with adversarial loss to enrich textures and bolster image color balancing quality. ResUnet [12] combines a UNet encoder–decoder structure with residual connections and atrous convolution modules to predict object boundaries, distance transforms, and reconstruct input images. This approach enhances the handling of class-imbalance issues, resulting in superior performance across various remote sensing datasets. DiffLL [11] proposes a wavelet-based conditional diffusion model for image enhancement. Initially, it applies wavelet transformation to convert images into the frequency domain, reducing spatial dimensionality and enhancing diffusion model efficiency and reconstruction accuracy. The model then conducts diffusion and denoising in the wavelet domain, coupled with a high-frequency recovery module to refine image details, which allows the enhanced image to maintain global contrast while achieving superior visual quality and restoration efficiency. WeatherDiffusion [12] presents an image restoration approach using a patch-based diffusion denoising model to restore images of arbitrary sizes, using smooth noise estimation to guide the sampling process and manage overlapping image patches. ABCNet [15] adopts a dual-path architecture, with a spatial path preserving rich spatial details and a context path capturing global context information. Through a feature aggregation module, the network effectively fuses these paths. Furthermore, it uses a linear attention mechanism to balance global information extraction with computational efficiency, achieving an optimal tradeoff between precision and computational load.

However, when applying these models to the ZHCBI remote sensing image color balancing task, several unresolved issues persist. During the network model's learning process on remote sensing images, the semantic consistency among similar category targets is not fully addressed. The end-to-end model proposed in this article places emphasis on the semantic association of similar targets, yielding promising results when applied to the ZHCBI remote sensing image color balancing task. As depicted in Fig. 1, TransGAN displays notable seam traces when color balancing the raw images for processing, attributable to significant color differences between adjacent image blocks. In contrast, our model achieves a more natural color balancing effect, effectively mitigating color differences between blocks and reducing sharp transitions caused by seam traces. Detailed comparative experimental results will be presented in Section IV.

C. Seamless Stitching Methods for Remote Sensing and Aerial Images

Seamless stitching of remote sensing and aerial images has long been a critical task in computer vision, with numerous researchers contributing to advancements in this domain [39]. One such method is the two-quarter circle (TQC)

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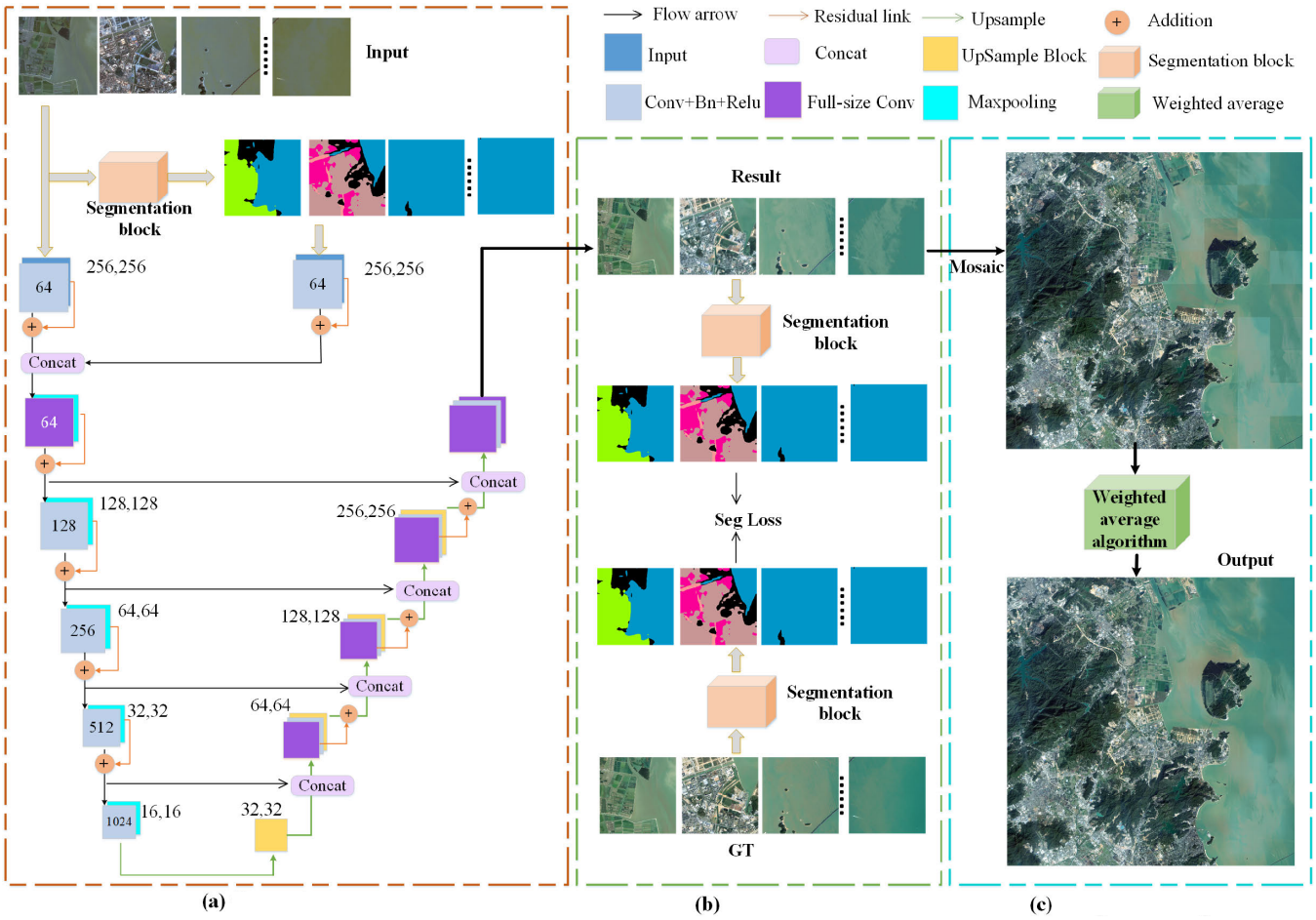


Fig. 2. Framework of our method with three key components, i.e., (a) color generator bolstered with a segmentation module, (b) segmentation style loss, tailored to enhance semantic association among targets within the same category, and (c) weighted averaging block specifically designed to tackle seams in stitched images.

algorithm [40], which calculates the gradient coefficient for each quarter circle. This technique not only ensures smooth transitions at image boundaries but also mitigates abrupt pixel variations at the stitching seams. Another approach [41] proposes an efficient color blending method for panoramic UAV image stitching. This technique centers around a selected UAV image, aligning overlapping images to this central reference. Subsequently, an optimal seam line is determined to remove ghosting artifacts. A weighted coordinate interpolation algorithm is then applied to compute the pixel color variations in the target image, further eliminating color discrepancies at the seam boundaries.

A superpixel-based stitching strategy [42] was also developed, which first uses adaptive as-natural-as-possible (AANAP) warping for image registration and alignment. Following this, an energy function based on superpixels is constructed, integrating color and texture information to identify the optimal stitching seam in the overlapping regions. Finally, a weighted coloring strategy is used to achieve seamless blending and eliminate visible seams. In addition, a UAV image stitching method based on optimal seam algorithms and a semi-projection baseline has been proposed [43]. This method first uses a differential matrix to preserve the color and structural information of the overlapping regions, followed by a minimum energy constraint search algorithm combined with a semi-projection baseline to maintain the original shapes of different regions, thereby enhancing the visual quality of the stitched images.

Moreover, a coarse-to-fine UAV image stitching algorithm [44] has been proposed, which uses GPS and IMU data to obtain positional and geometric information for coarse alignment. This is followed by the application of a multilayer perceptron (MLP) for color correction, achieving an optimal balance between speed and accuracy.

III. Method

A. Overall Framework

Fig. 2 outlines the comprehensive framework of our method, comprising the generator, segmentation style loss, and a weighted averaging algorithm for postprocessing. Our framework consists of three components: the generator, segmentation style loss, and weighted averaging block. The generator, depicted in Fig. 2(a), accepts multiple 256 × 256 image blocks as input, processing them to generate uniformly colored output images. This generator efficiently

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normalizes the colors of diverse remote sensing images, restoring them to appear natural and aligning with human visual perception. In addition, a segmentation module is incorporated to enhance the semantic association among targets of the same category, thereby augmenting the color balancing effect of remote sensing images.

The second part concerns the segmentation style loss, as depicted in Fig. 2(b). Both the generated images and their corresponding ground-truth counterparts undergo processing via the segmentation module to generate semantic category segmentation maps. Subsequently, a segmentation style loss function is designed to quantify the disparities between these maps, thereby guiding the training of the generator. The aim of this loss function is to improve the generator's focus on targets within the same category, thereby ensuring color consistency among adjacent image blocks. Consequently, this helps diminish color discrepancies and leads to the creation of more natural and precise images.

The third component, illustrated in Fig. 2(c), is the weighted averaging module, which addresses the challenge of visible seams following image block stitching. We have developed an efficient and practical weighted averaging postprocessing algorithm that swiftly eliminates horizontal and vertical stitching lines in the stitched images, effectively mitigating potential seam issues at the code level. This algorithm operates autonomously, eliminating the need for manual intervention, thereby minimizing potential errors and ensuring broad applicability across various model outputs.

B. Generator

The generator, illustrated in Fig. 2(a), comprises three main components: a segmentation module, an encoder, and a decoder. The input consists of a series of 256 × 256 ZHCBI remote sensing image blocks, initially processed by a semantic segmentation module to generate corresponding semantic segmentation maps. These input images and maps then traverse through two sets of convolutional residual blocks, progressively increasing the feature map count from three to 64. The resulting outputs are channel-stacked to boost the feature map count to 128. Subsequently, a full-size convolutional residual module further enhances these maps, reducing the channel count from 128 to 64 to improve the model's feature learning capability. This processed information is then fed into the encoder, which captures low-frequency pixel information, aiding in recognizing key features such as object shape, size, and texture.

The encoder is structured into four groups, each comprising convolutional residual blocks and a max-pooling layer. Across these groups, the number of input feature maps gradually increases from 64 to 1024, while the max-pooling layers progressively downsample the input feature maps from 256 × 256–16 × 16. Upon completion of the encoder section, it outputs 1024 feature maps, each sized 16 × 16.

The decoder consists of five groups, marked by green arrows in Fig. 2(a), representing the upsampling operations. Features from corresponding levels in the encoder are concatenated with those at the same levels of the decoder to retain more texture details. Subsequently, through full-size convolution, batch normalization (BN), and ReLU operations, the model's training stability is ensured, preserving more semantic information. Post upsampling, the feature maps are restored to match the size of the input images, yielding the final output image.

For the semantic segmentation module, we selected WiCoNet [26], a context-based transformer structure semantic-aware model known for its excellent performance in semantic segmentation of remote sensing images, particularly on the high-resolution Gaofen image dataset (GID). We trained WiCoNet on the GID dataset and retained the best testing model for integration into our color balancing task's segmentation module, as depicted in Fig. 2(a). Importantly, the segmentation module used in this study serves solely to extract semantic feature information from images and does not directly partake in the model's training process.

C. Loss Functions

The loss functions used in this article consist of segmentation style loss and pixel-level loss, designed to effectively enhance the quality of the generated images following color balancing processing. As illustrated in Fig. 2(b), the segmentation style loss leverages the segmentation module to conduct semantic segmentation on both the network-generated and real results, thereby obtaining corresponding semantic segmentation maps for each. This loss function quantifies the disparity between the pixel classifications of the generated image and the ground-truth values, thereby aiding in improving the quality of the generated images concerning color and edge handling. Furthermore, it encourages uniform color generation for targets of the same category within the images. To calculate the discrepancy between these two semantic segmentation maps, a modified version of cross-entropy loss is used

$$t_{\text{result}}, t_{\text{GT}} = F(G(I_{\text{Input}}), S), F(I_{\text{GT}}, S). \tag{1}$$

Among them, $G(I_{\text{Input}})$ represents the network-generated result, and the function F encompasses preprocessing operations such as scaling, padding, and cropping of the image. I_{Input} represents the ground truth of the ith pixel in the image, and I_{GT} represents the input for the real image. Before semantic segmentation extraction, both the generated results and the real images need to undergo identical scaling, padding, and cropping operations

$$t^C_{\text{result}}, t^C_{\text{GT}} = \text{Color2index}[t_{\text{result}}, t_{\text{GT}}], \quad C \in [0, 16]. \tag{2}$$

Color2index refers to the process of converting the categories in the semantic segmentation map into indices. In this article, the number of classes for remote sensing image semantic segmentation is set to 16, with index 0 indicating an unrecognized category. The formula for the segmentation style loss is as follows:

$$L_{\text{seg}} = -\frac{1}{N}\sum_{i=1}^{N} w_i \log \frac{\exp\left(P_{\text{input}}\left(i, t^C_{\text{GT}}\right)\right)}{\sum_{j=1}^{C}\exp\left(P_{\text{input}}\left(i, t^j_{\text{result}}\right)\right)} \tag{3}$$

where N represents the total number of pixels, which is the product of the image's height H and width W; C denotes the number of categories, ranging from $[0, 16]$; $P_{\text{input}}(i, j)$

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Algorithm 1 Weighted Average Algorithm

Input : $\bar{E}_{ij}, i, j \in [1, n]$, E_{ij} is on image with a size of 256×256. i represents the ith row image, and j represents the jth column image

Output: The complete remote sensing image without stitching lines $\mathrm{S} = \hat{S}$

for $j \rightarrow 1$ **to** n **do**
  for $i \rightarrow 1$ **to** $n-1$ **do**
    $t_i = E_{ij} \oplus E_{(i+1)j}$
    if $i = 0$ **then**
      $t_j = t_i$
    end
    else $\iota_j = \iota_j \oplus \iota_i$;
  end
  $\hat{E}_j = t_j$
end
for $j \leftarrow 1$ **to** n **do**
  for $i \leftarrow 1$ **to** $n-1$ **do**
    $\hat{E}_j = W_1 \times E_{ij} + W_2 \times E_{(i+1)j}$
  end
  $\mathrm{E}_j = \hat{E}_j$
end
for $j \leftarrow 1$ **to** $n-1$ **do**
  $\hat{S} = E_j \oplus E_{j+1}$
end
for $j \leftarrow 1$ **to** $n-1$ **do**
  $\hat{S} = W_1 \times E_j + W_2 \times E_{j+1}$
end
Return $\mathrm{S} = \hat{S}$

indicates the predicted probability that the ith pixel belongs to the jth category; t_{GT}^{C} denotes the true category of the ith pixel; and w_i is an optional category weight, which can be used to emphasize the importance of a specific category and can be set at initialization. Notably, unrecognized categories, where $C = 0$, are disregarded when calculating the segmentation style loss.

The pixel-level loss formula is as follows:

$$L_{\mathrm{pix}} = \|G(I_{\mathrm{input}}) - I_{\mathrm{GT}}\|_1 \tag{4}$$

where $G(I_{\mathrm{Input}})$ represents the network-generated results, and I_{GT} denotes the input of the real images.

The proposed multilevel cascaded loss function is as follows:

$$L_{\mathrm{dl}} = \lambda_1 L_{\mathrm{seg}} + \lambda_2 L_{\mathrm{pix}} \tag{5}$$

where λ_1, λ_2 represent the weight coefficients for the segmentation style loss and the pixel-level loss, respectively.

D. Weighted Averaging Postprocessing Algorithm

Extensive experimentation uncovered a notable issue with visible stitching lines in the full remote sensing images assembled from blocks (256×256) generated by the generator. Traditional median and Gaussian filters [45], [46] proved inadequate in resolving this issue. Moreover, while SIFT image stitching techniques [47] could conduct feature point matching and stitching, the resulting images still displayed evident stitching lines.

To tackle this challenge, we introduce a weighted averaging algorithm aimed at effectively eliminating horizontal and vertical stitching lines in remote sensing images. This algorithm achieves a visually seamless effect conducive to practical applications. The algorithm proceeds as follows.

First, as it is not possible to simultaneously eliminate horizontal and vertical stitching lines in remote sensing images, we opt to first address the horizontal lines (the method for vertical lines is similar). The generated blocks, E_{ij}, are stitched together column by column, resulting in several single-column images, $\hat{E}_j$, each with horizontal stitching marks.

The entire remote sensing image is divided into $i \times j$ blocks, where i denotes the row and j the column. Each block E_{ij} is $W \times H$ pixels, with overlapping edges of R rows or C columns. These columns of blocks are then vertically stitched to form single-column images, $\hat{E}_j$, with horizontal lines.

Second, weights P_1 and P_2 are set, ranging from $[0, R]$, where P_1 decreases from $\mathrm{MaxP}_1\%$ to $\mathrm{MinP}_1\%$, and P_2 increases from $\mathrm{MinP}_2\%$ to $\mathrm{MaxP}_2\%$. The overlapping R rows of pixels between adjacent blocks E_{ij} and $E_{(i+1)j}$ are averaged using the formula $P_1 \times E_{ij} + P_2 \times E_{(i+1)j}$, and the averaged pixels are assigned to their corresponding positions in $\hat{E}_j$. This process is repeated across all $\hat{E}_j$ to eliminate the horizontal stitching marks, resulting in single-column images without horizontal lines.

Third, these processed single-column images E_j are then reassembled, forming a high-resolution remote sensing image with vertical stitching marks, $\hat{S}$.

Finally, adjacent columns E_j and $E_{(j+1)}$ are processed using weighted averaging, using the formula $P_1 \times E_j + P_2 \times E_{(j+1)}$. The pixels resulting from this averaging are assigned to their corresponding positions in $\hat{S}$ to complete the vertical averaging operation. This concludes the weighted averaging process, resulting in a seamless remote sensing image S, which serves as the final output of our framework, as depicted in the output of Fig. 2(c).

Our weighted averaging algorithm effectively eliminates stitching lines in remote sensing images at the code level. It is fast and requires no manual intervention. Furthermore, it is highly applicable, suitable for processing outputs from various models.

IV. Experiments

A. Datasets and Evaluation Metrics

In this experiment, we used two datasets, including our proposed ZHCBI dataset and a dataset derived from the GID dataset [48].

First, we selected land–ocean images captured by the Gaofen-1 satellite over the Zhuhai area, named ZHCBI, to serve as a new benchmark image for remote sensing color balancing processing. This image with dimensions of $17\,900 \times 13\,800$ pixels consists of multiple regional images compiled into a complete high-resolution satellite image. As shown in Fig. 3, five blocks of images were selected for training and another three blocks for testing. To fully use

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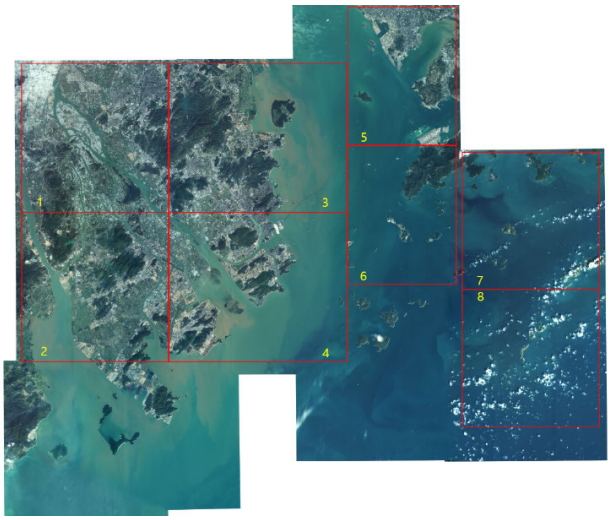


Fig. 3. Original image used for dataset creation. This true-color satellite image was manually colored by professionals in the field, ensuring data accuracy and practicality. Eight areas were selected from this image to serve as the ground-truth samples for the training dataset, marked by red borders and assigned numeric identifiers.

GPUs for training, these images were further cropped into several 256 × 256 subimages. The dataset is accessible via http://39.98.109.195:1000/share/xTVn60gY.

As depicted in Fig. 3, the red borders indicate dataset images extracted from the high-resolution satellite image with dimensions of 17 900 × 13 800 pixels. To ensure robust generalizability of the model posttraining, we captured a sufficient number of land and ocean images for both training and testing. After segmentation and cropping using OpenCV, areas 1, 4, 5, 6, and 8, marked in yellow font, were designated as the training set, containing a total of 26 880 images, including 8960 training images and their corresponding 8960 labeled images, along with 8960 semantic segmentation images of these training images. Similarly, areas 2, 3, and 7 were allocated as the validation image set, comprising 15 360 validation images, including 5120 to-be-validated images along with their 5120 labeled images and 5120 corresponding semantic segmentation images. Moreover, during testing, validation data may be required and can be selected from the validation set as needed. All the training, validation, and test images were cropped to 256 × 256 pixels, ensuring no overlap between the training and validation sets.

Second, as shown in Fig. 4, we selected four remote sensing images from the GID dataset [48] for testing. This dataset includes a true-color remote sensing image with dimensions of 5600 × 5600 pixels, divided into four regions. Each region is marked with a red border and labeled as 1, 2, 3, and 4. These regions were cropped into 256 × 256 subimages, resulting in a total of 400 test images. Without retraining the models, we directly tested each of the pretrained optimal models on this dataset.

To objectively evaluate the performance of our model and other comparative methods, we used two full-reference metrics: peak signal-to-noise ratio (PSNR) [49] and structural similarity index (SSIM) [50], to assess the result images. These metrics reflect the closeness of the generated results to the true labels. A higher PSNR value indicates closer content information, while a higher SSIM value indicates greater similarity in structure and texture information.

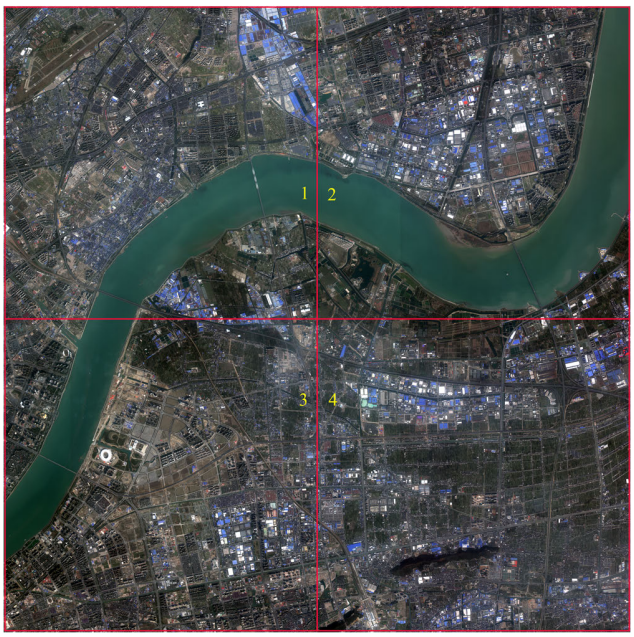


Fig. 4. Remote sensing dataset collected by the GID dataset [48]. It has dimensions of 5600 × 5600 pixels, with four regions selected as ground-truth samples for the test dataset. These regions are marked with red borders and assigned numerical identifiers.

B. Experimental Settings

All the experiments in this article were implemented using the PyTorch framework. The experiments were conducted in a Windows environment, using an Intel[1] Core[2] i9-10 900 K CPU at 3.70 GHz with 128 GB of memory, along with two NVIDIA GeForce RTX 3090 GPUs. During the model training phase, we used the Adam optimizer, with $\beta_1 = 0.9$, $\beta_2 = 0.999$, and an initial learning rate of 0.0001. The learning rate was updated every five epochs at 80% of the previous value until the end of the experiment. The training was set at 150 epochs, with a batch size of 16, and the best-quality model was saved during training. For the loss function configuration, the training weight for segmentation style loss was set at 100.0, and for pixel-level loss, it was set at 1.0. The entire training process took approximately 32 h.

In addition, we used WiCoNet [26] as the semantic segmentation module (the segmentation block in Fig. 2). It was trained on the high-resolution GID for 50 rounds, which took about 27.5 h in total. After testing the model, we found that the best results were obtained at the 40 epochs of training; this optimal model was then used as the segmentation module for the color balancing task.

To better perform qualitative and quantitative analyses, we compared the color balancing effects of our model with several recently proposed models [7], [8], [9], [10], [11], [12], [15], [16], [17].

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[2]Trademarked.

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TABLE I
COLOR BALANCING RESULTS ON THE ZHCBI DATASET OBTAINED BY OUR MODEL COMPARED WITH OTHER EXISTING MODELS. (NOTE: ↑ INDICATES THAT HIGHER IS BETTER, WHEREAS ↓ INDICATES THAT LOWER IS BETTER)

Methods	Venue	PSNR↑	SSIM↑
ResUnet [7]	ISPRS'2020	22.08	0.7546
TransGAN [8]	NeurIPS'2021	21.78	0.7551
MANet [16]	TGRS'2021	22.26	0.7335
ABCNet [15]	ISPRS'2021	21.22	0.6869
UNetFormer [17]	ISPRS'2022	21.76	0.6803
LLFlow [9]	AAAI'2022	20.86	0.7826
GFMT [10]	TIP'2023	21.38	0.7364
DiffLL [11]	TOG'2023	22.37	0.7843
WeatherDiff [12]	TPAMI'2023	22.25	0.8007
SECBNet	Our work	**23.64**	0.7798

TABLE II
COLOR BALANCING RESULTS ON THE GID DATASET OBTAINED BY OUR MODEL COMPARED WITH OTHER EXISTING MODELS. (NOTE: ↑ INDICATES THAT HIGHER IS BETTER, WHEREAS ↓ INDICATES THAT LOWER IS BETTER)

Methods	Venue	PSNR↑	SSIM↑
ResUnet [7]	ISPRS'2020	21.36	0.8020
TransGAN [8]	NeurIPS'2021	20.23	0.8887
MANet [16]	TGRS'2021	19.97	0.8354
ABCNet [15]	ISPRS'2021	20.73	0.7225
UNetFormer [17]	ISPRS'2022	20.91	0.7258
LLFlow [9]	AAAI'2022	20.04	0.8581
GFMT [10]	TIP'2023	21.30	0.7318
DiffLL [11]	TOG'2023	20.97	0.8639
WeatherDiff [12]	TPAMI'2023	19.39	0.8269
SECBNet	Our work	**22.63**	0.8187

C. Quantitative Analysis

Tables I and II present a comparison of each testing method based on key metrics for the ZHCBI and GID datasets, specifically focusing on PSNR and SSIM. Our model attained the highest PSNR value, suggesting a strong emphasis on semantic consistency within categories to improve image generation quality. ResUnet [7] stands out for its moderate parameter size and faster training speed. However, its single network structure restricts its ability to grasp complex textures and color information effectively. Conversely, the TransGAN [8] model lacks adequate consideration of semantic associations between categories. The UNetFormer's [17] global transformer decoder shows insufficient capability in restoring global features such as color and texture, as evidenced by its relatively low PSNR and SSIM values. While the MANet's [16] multiattention module successfully integrates multiscale features, its performance in restoring texture details remains suboptimal. Similarly, the GFMT module in [10] struggled to focus on degraded color information, resulting in inferior contrast and texture detail preservation. DiffLL [11] and WeatherDiff [12] have slightly higher SSIM metrics than our model, but our model still holds a leading position in PSNR metrics. In contrast, our method achieved the best PSNR value overall, demonstrating its effectiveness in restoring color information. Its SSIM value also highlights its relative advantage in preserving texture structure, further validating the efficiency of our method in the remote sensing image color balancing task.

D. Visualization Analysis

In Fig. 5, we present qualitative visual comparison results from selected methods. The first three rows correspond to the ZHCBI test set, while the subsequent four rows belong to the GID dataset test set. We selected 245 images from the ZHCBI test set and 400 images from the GID dataset, each sized 256 × 256 from the test set, which were stitched together to form a complete 2560 × 2560 remote sensing image. From Fig. 5, it is evident that GFMT model exhibits notable instability in categories such as oceans, displaying color biases and unnatural colors. This may be attributed to its decoupling and fusion transformer module, which only conducts local learning on image blocks, thus failing to consider the low-frequency correlations between blocks. While ResUnet and TransGAN manage to generate images with somewhat consistent low-frequency color information, they have not fully addressed the inconsistency in correlation across different color channels, resulting in color biases in some stitched blocks. LLFlow shows better color saturation, reducing color distortion, but there are instances where individual blocks differ significantly in color from surrounding blocks, likely due to color overlapping or sharp noises in its color mapping module. MANet demonstrates better color transitions between adjacent images; however, it still exhibits overly bright areas in high-luminance regions. UNetFormer shows significant blank gaps at the seams, indicating its relatively poor ability to restore image boundaries.

It is clear that our method exhibits a more natural visual effect compared with others, particularly in color transitions and contrast. In images featuring objects within the same category (such as oceans, forests, and buildings), the stitched images display smoother color transitions, and the generated block images appear more natural. These stitched blocks effectively reduce the visual traces of stitching lines, further diminishing the color differences between adjacent blocks, thus achieving the best visual quality. However, as evident from Fig. 5, even our model is not entirely immune to stitching traces. To address this issue, we introduced the weighted averaging algorithm in Section III-D, through which we successfully eliminated the traces produced by stitching adjacent blocks. Our weighted averaging algorithm exhibits strong generality and generalization capabilities, making it suitable for processing outputs from any model. Fig. 6 illustrates this effect: the first row shows images before processing, while the

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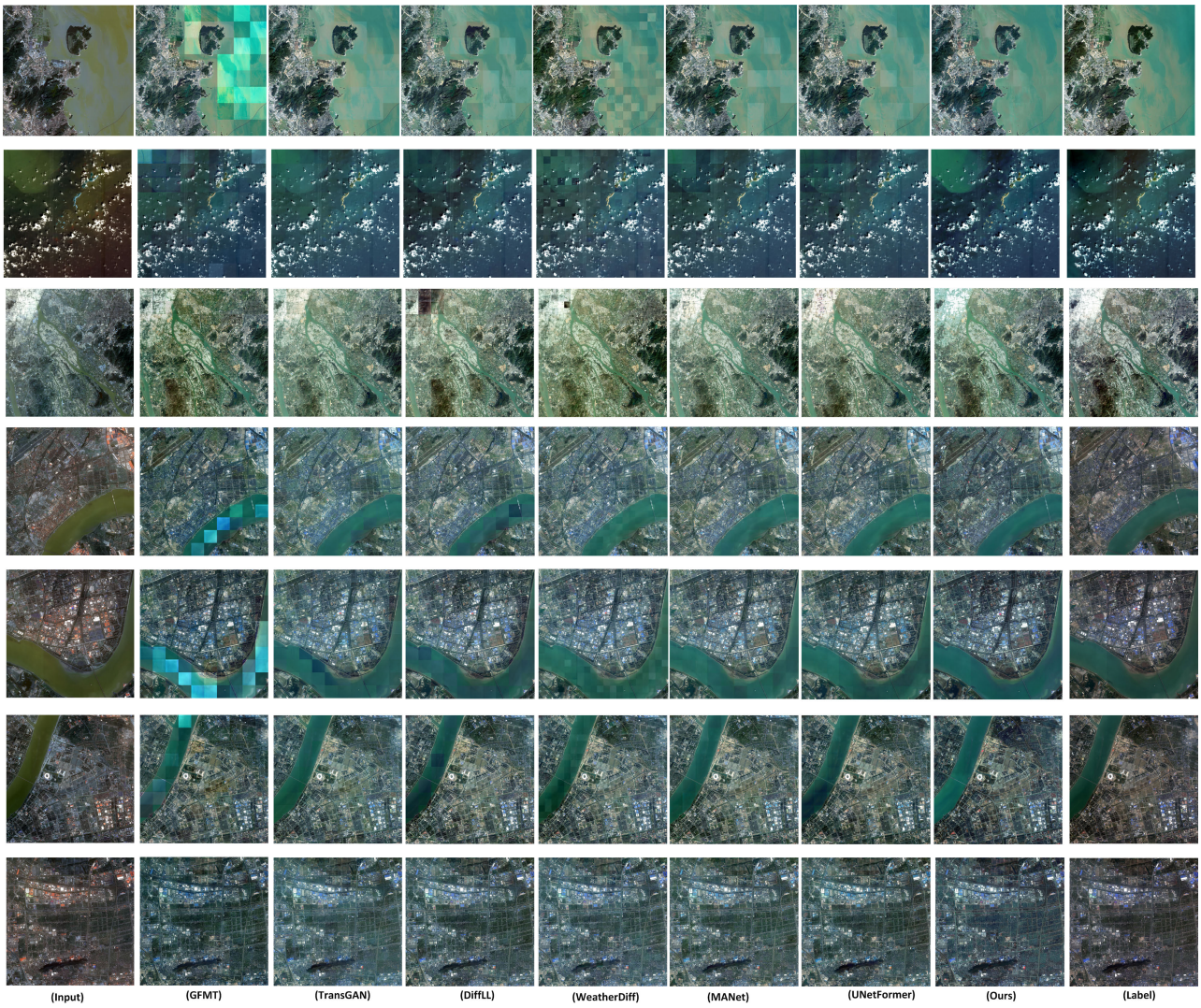


Fig. 5. Visual comparison of the results after color balancing the test images of the ZHCBI and GID datasets by our model and several advanced models. The first three rows represent the ZHCBI dataset, while the subsequent four rows represent GID dataset. The figure above denotes the input images, GFMT, TransGAN, DiffLL, WeatherDiff, MANet, UNetFormer, our model, and labels, respectively.

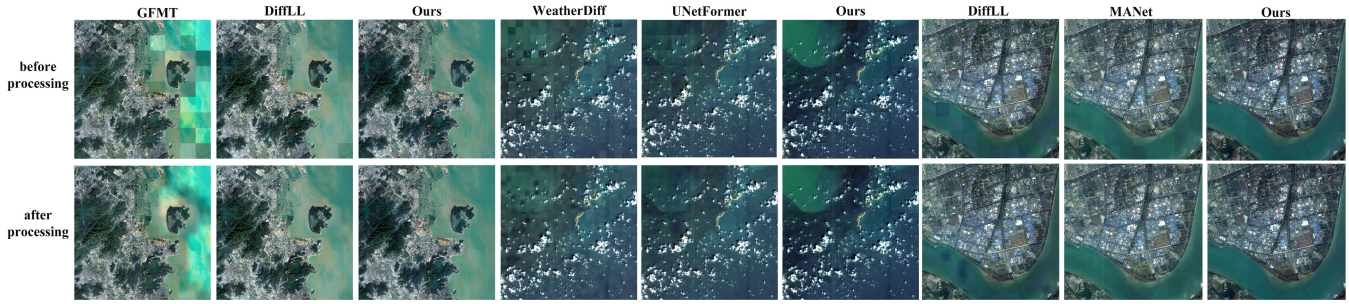


Fig. 6. Visual comparison of the effects of our weighted averaging algorithm on the complete remote sensing images formed by stitching blocks from different model outputs (top row: images before processing, bottom row: images after processing).

second row displays images after processing with the weighted averaging algorithm. We selected postprocessed result images with better visual quality from the ZHCBI and GID datasets, respectively. Through this algorithm, we successfully eliminated all traces of stitching that arose from joining blocks.

E. Effectiveness of the Weighted Averaging Algorithm

As displayed in Table III, the algorithm effectively eliminates visual stitching traces. Several models, including ours, have seen improvements in PSNR, indicating that the weighted averaging algorithm can not only eliminate stitching seams but also enhance image quality to a certain extent. However, we observed decreases in PSNR and SSIM for GFMT; GFMT's PSNR decreased by 0.2 dB, which may be attributed to significant color discrepancies between blocks. After the weighted averaging algorithm, our model's PSNR increased by 0.41 dB, and our model still maintains the highest PSNR value, showcasing its robustness and generalizability. Overall,

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TABLE III

RESULTS OBTAINED BY OUR ALGORITHM ON THE ZHCBI DATASET WERE COMPARED WITH OTHER MODELS. (NOTE: ↑ INDICATES THAT HIGHER IS BETTER)

Methods	**Pre-Processing PSNR↑**	**Post-Processing PSNR↑**	**Pre-Processing SSIM↑**	**Post-Processing SSIM↑**
image types	All Test Images	All Test Images	All Test Images	All Test Images
ResUnet	22.08	22.36	0.7546	0.7732
TransGAN	21.78	22.12	0.7551	0.7678
MANet	22.26	22.87	0.7335	0.7533
ABCNet	21.22	21.68	0.6869	0.7091
UNetFormer	21.76	22.18	0.6803	0.7040
LLFlow	20.86	21.31	0.7826	0.7978
GFMT	21.38	21.18	0.7364	0.7441
DiffLL	22.37	22.60	0.7843	0.7921
WeatherDiff	22.25	22.36	0.8007	0.8065
Ours	**23.23**	**23.64**	0.7683	0.7798

TABLE IV

RESULTS OBTAINED BY OUR ALGORITHM ON THE GID DATASET WERE COMPARED WITH OTHER MODELS. (NOTE: ↑ INDICATES THAT HIGHER IS BETTER)

Methods	**Pre-Processing PSNR↑**	**Post-Processing PSNR↑**	**Pre-Processing SSIM↑**	**Post-Processing SSIM↑**
image types	All Test Images	All Test Images	All Test Images	All Test Images
ResUnet	21.36	21.65	0.8020	0.8100
TransGAN	20.23	20.25	0.8887	0.8963
MANet	19.97	20.18	0.8354	0.8407
ABCNet	20.73	20.89	0.7225	0.7332
UNetFormer	20,91	21.00	0.7258	0.7283
LLFlow	20.04	20.43	0.8581	0.8562
GFMT	21.30	21.11	0.7318	0.7413
DiffLL	20.97	21.18	0.8639	0.8756
WeatherDiff	19.39	19.67	0.8269	0.8474
Ours	**22.56**	**22.63**	0.8174	0.8187

the visual results of our algorithm more closely adhere to human visual esthetic standards and can improve image quality to a certain extent.

In summary, for more effective utilization of model-generated remote sensing images in research or practical applications, it is recommended to use our proposed weighted averaging algorithm to produce visually seamless satellite images.

The results in Table IV also demonstrate the robustness and generalization capability of our weighted averaging algorithm. When applied to the GID dataset, the weighted averaging still enhances the PSNR and SSIM values of the remote sensing images.

F. Discussion of Semantic Segmentation Block Convolutional Depth

As depicted in Fig. 7, the diagram illustrates the depth stratification of feature extraction in semantic segmentation maps across different convolutional layers. The diagram begins

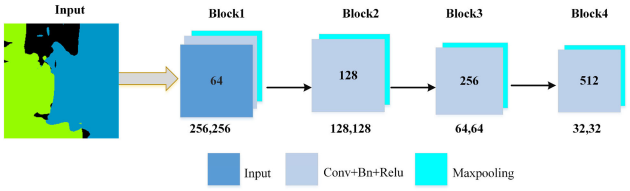


Fig. 7. Depth stratification of semantic segmentation maps at different convolutional levels, from Blocks 1 to 4. As the images progress through these blocks, the number of feature maps increases from 64 to 128, and then to 512, while the resolution of the images gradually decreases from 256 × 256 to 128 × 128, eventually reaching 32 × 32.

TABLE V

USE OF THE SEGMENTATION MODULE AND EFFECT OF DIFFERENT DOWNSAMPLING DEPTHS OF THE SEGMENTATION MODULE ON OUR MODEL'S GENERATED RESULTS. (NOTE: ↑ INDICATES THAT HIGHER IS BETTER)

Metrics	PSNR↑	SSIM↑
Image types	All Test Images	All Test Images
Ours-w/o Seg	22.01	0.7524
Ours-Seg64	**23.23**	0.7683
Ours-Seg128	22.75	0.7729
Ours-Seg256	22.80	0.7738
Ours-Seg512	22.94	0.7776

with "Input," representing the semantic segmentation map input, followed by four blocks (Blocks 1–4), symbolizing feature maps processed through successive convolution layers. The numbers atop each block (64, 128, 256, 512) denote the depth of the feature maps, indicating the number of channels, while the corresponding numbers at the bottom (e.g., 256 × 256, 128 × 128) represent the spatial resolution of the feature maps at that layer. Our aim is to analyze the impact of feature map depth following each convolution level on model performance, aiming to ascertain the optimal depth for feature extraction.

As outlined in Table V, we conducted ablation experiments comparing the use of the segmentation module and its absence. The experimental setup adhered to the optimal configuration identified in this study, and the output on the test set was compared using PSNR and SSIM values. The results reveal that using the Seg64 segmentation module yielded the highest PSNR at 23.23 dB. Furthermore, it was observed that as the depth of the segmentation module increased, the PSNR performance of the test results gradually declined. In addition, the absence of a segmentation module resulted in the lowest PSNR at 22.01 dB. In comparison to the approach without a segmentation module, our optimal model enhanced the PSNR by 1.22 dB and the SSIM by 0.0159. In addition, we also analyzed the repair effectiveness of the model when the segmentation module is added or removed, as shown in Fig. 8.

Ablation experiments were conducted by selecting different overlapping pixels for weighted averaging, as shown in Fig. 9. By comparing Algorithms (1)–(3), it can be observed in the regions marked by the red boxes that Algorithm (1), with a 12-pixel overlap, eliminates some stitching seams but introduces new ones. Algorithm (2) applies Gaussian filtering, where the farther a pixel is from the center, the smaller its weight, and pixels closer to the center have larger weights.

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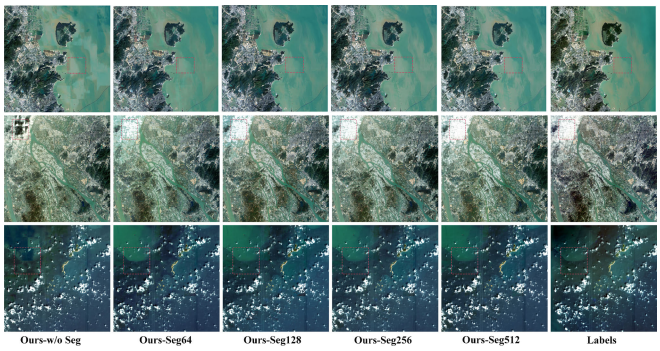


Fig. 8. Results of the ablation experiment (red box: the area in the image with considerable difference).

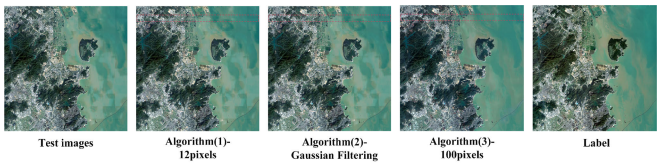


Fig. 9. Results of the weighted averaging algorithm using different overlapping pixel values are shown on the test images (red boxes: areas with significant differences in the image).

TABLE VI
PERFORMANCE COMPARISON AMONG TEN METHODS ON THE ZHCBI DATASET

Methods	Parameters(MB)	Memory(MB)	Time(s)
ResUnet	179.5	360.9	23.3
TransGAN	1290.6	1550.9	69.1
MANet	136.8	268.8	25.4
ABCNet	50.9	74.4	23.1
UNetFormer	44.7	59.8	22.8
LLFlow	20.7	58.4	72.4
GFMT	120.5	302	31.0
DiffLL	84.2	231.1	91.4
WeatherDiff	326.6	1424.1	2442.7
SECBNet	180.4	378.5	27.6

However, notable stitching seams still appear. Algorithm (3), which overlaps 100 pixels, is the method ultimately adopted, as it visually eliminates the stitching seams completely.

G. Performance Comparison of Different Methods on the ZHCBI Dataset

As displayed in Table VI, we tested the ZHCBI test set on a single NVIDIA GTX 3060 GPU using an input size of 256 × 256 and reported the parameters, memory usage, and testing time. It can be seen that our segmentation-enhanced color balance network (SECBNet) method has a testing time of 27.6 s, only 4.8 s slower than the fastest method, UNetFormer. Although UNetFormer and ABCNet have shorter testing times, considering their smaller parameter sizes and memory usage, our method still demonstrates an excellent balance between computational time and model performance. In addition, our parameter count is comparable to ResUnet and MANet, indicating an effective model simplification without significantly compromising performance. In summary, our method excels in balancing model complexity and resource consumption, especially in inference speed and memory efficiency, showing advantages over some of the more complex models.

V. CONCLUSION

This article presented a novel satellite image color balance work with the following contributions. First, an open access color balancing dataset, named ZHCBI, was created and published to overcome the scarcity of data in optical satellite image processing research fields, and we constructed and publicly released an optical remote sensing image. This dataset uses the Gaofen-1 satellite images over the Zhuhai area as base data, with color balancing labels obtained through expert manual adjustment. Second, a dual-branch color balancing network based on semantic segmentation information was presented to enhance the semantic consistency between color balanced images and maintain global color consistency for the same categories across different images. In addition, a joint loss function that incorporates segmentation style loss was designed to guide model learning. Finally, a postprocessing algorithm based on segmented weighted averaging was proposed to eliminate the seams between different processed blocks. The proposed method was compared with nine other methods, i.e., ResUnet, TransGAN, MANet, ABCNet, UNetFormer, LLFlow, GFMT, DiffLL, and WeatherDiff on the ZHCBI dataset. Our method not only demonstrated better visual color consistency but also achieved higher PSNR values. In the ZHCBI dataset color balancing task, our method achieved a PSNR value of 23.64 dB, which is 1.27 dB higher than the WeatherDiff method. In future work, we plan to focus on enhancing the network's ability to restore texture structure and further explore into remote sensing image color balancing.

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